

(Group H)

Short Use Case Template for COMP3297: BetaTrax.

BetaTrax Use Case

UC ID and Name:	BT-UC1 Handle product registration		
Primary Actor:	Product Owner	Supporting Actors:	User Authorization System
Trigger:	A product participates in the beta testing program and is tracked in BetaTrax.		
Description:	The Product Owner registers a product in BetaTrax and associates the product with its owner and developer so that defect reports can be received and managed.		
Preconditions:	PRE-1: Product Owner is authenticated. PRE-2: The product is not already registered in BetaTrax. PRE-3: The Product Owner is not already assigned as the owner to another product in BetaTrax. PRE-4: Each selected Developer is authenticated to BetaTrax and is not already assigned to another product.		
Postconditions:	POST-1: A new Product record exists with a unique Product ID. POST-2: The Product Owner and Developers are linked to the Product.		
Basic Flow:	1. Product Owner requests to register a new product and provides Product ID. 2. Product Owner provides the list of Developers to associate with the product. 3. System validates that Product ID is unique and that the owner and developers are not assigned to another product. 4. System creates the Product record and stores owner and developer associations.		
Alternative Flows:	3a. Product ID already exists: 1. System rejects the request and indicates the Product ID is already registered. 3b. Owner or Developer already assigned to another product: 1. System rejects the request and identifies the conflicting user(s). 2. Product Owner revises the request or cancels.		

UC ID and Name:	BT-UC2 Submit Defect Report		
Primary Actor:	Beta Tester	Supporting Actors:	Beta UI, Email Service
Trigger:	Tester report a defect from the beta product UI after encountering an issue.		
Description:	A tester submits a defect report for a specific product and beta version. BetaTrax assigns it an ID, sets the status to New, and stores it for Product Owner evaluation.		
Preconditions:	PRE-1: Tester has a valid Tester ID. PRE-2: The product is registered in BetaTrax. PRE-3: The beta product can connect to the BetaTrax API.		
Postconditions:	POST-1: A new Defect Report is stored with status = "New" and a received date/time. POST-2: The report is associated with the specified Product ID and version.		
Basic Flow:	1. Tester enters the defect report details: Product ID and version, title, description, steps to reproduce, Tester ID, and optionally an email address for notifications.		

	2. Beta Product submits the report to BetaTrax. 3. System validates required fields and that the Product ID exists. 4. System assigns a unique Report ID, records the received date/time, and sets status to “New”. 5. System stores the report and returns the Report ID to the Beta Product.
Alternative Flows:	3a. Unknown Product ID or invalid version: 1. System rejects the submission and returns an error. 2. Tester corrects the product/version and resubmits.

UC ID and Name:	BT-UC3 Evaluate New Defect Report		
Primary Actor:	Product Owner	Supporting Actors:	Email Service, Backlog Tool
Trigger:	A Product Owner reviews a newly submitted defect report with status = “New”.		
Description:	The Product Owner evaluates a New defect report and decides to accept, Reject, or mark it as a Duplicate. If accepted, severity and priority are recorded and a Product Backlog item reference is stored; the report status becomes Open.		
Preconditions:	PRE-1: Product Owner is authenticated. PRE-2: The defect report exists and has status = “New”. PRE-3: The defect report belongs to the Product Owner’s product.		
Postconditions:	POST-1: The defect report status is updated to one of: Open, Rejected, Duplicate. POST-2: If accepted, Severity and Priority are recorded and a Backlog item reference is stored. POST-3: If duplicate, the report is linked to the report it duplicates. POST-4: If the tester provided an email address, a status-change notification email is sent.		
Basic Flow:	1. Product Owner requests the list of New defect reports for their product. 2. System displays the list; Product Owner selects a report. 3. Product Owner chooses “Accept” and records Severity, Priority, and the Product Backlog item reference. 4. System records these values, sets report status to “Open”, and stores the decision timestamp. 5. System sends a status-change email notification to the tester if an email address was provided.		
Alternative Flows:	3a. Reject report: 1. Product Owner chooses “Reject”. 2. System sets status to “Rejected” and stores the decision timestamp. 3. System emails the tester if an email address was provided. 3b. Mark report as duplicate: 1. Product Owner chooses “Duplicate” and selects the existing report it duplicates. 2. System sets status to “Duplicate”. 3. System emails the tester if an email address was provided.		

UC ID and Name:	BT-UC4 Developer View and Take Defect		
Primary Actor:	Developer	Supporting Actors:	User Authorization System, Email Service
Trigger:	An Open defect is selected for work from the Backlog into the upcoming sprint, and a developer chooses to take the ownership.		
Description:	A developer accepts responsibility for resolving an accepted defect report. BetaTrax records the status change and notifies the tester by email if applicable.		

Preconditions:	1. PRE-1: The defect report exists and its status is Open. 2. PRE-2: The developer is authenticated and is part of the product's developer team.
Postconditions:	1. POST-1: The defect report is linked to the responsible developer. 2. POST-2: The defect status is updated to Assigned. 3. POST-3: The assignment timestamp/audit trail is recorded. 4. POST-4: The system sends an email notification to the tester to inform the status change if applicable.
Basic Flow:	1. Developer views an Open defect report detail. 2. Developer takes responsibility for resolving the defect. 3. System records the developer assignment on the defect report. 4. System changes the defect status from Open to Assigned. 5. System confirms the assignment and shows the updated status/assignee. 6. System sends a notification email to the tester if an email address is provided.
Alternative Flows:	3a. Developer declines/cancels taking ownership: 1. System does not record any assignments. 2. Use case ends (defect remains Open, no assignee recorded).

UC ID and Name:	BT-UC5 Developer Update Defect as Fixed or Cannot Reproduce		
Primary Actor:	Developer	Supporting Actors:	User Authorization System, Email Service
Trigger:	The assigned developer finishes investigating the defect and either implements a fix, or cannot reproduce the reported defect.		
Description:	The assigned developer updates the defect report after investigation. If a correction is implemented, the developer marks the defect as Fixed. If the developer cannot reproduce the issue, the developer marks it as Cannot reproduce. BetaTrax records the status change and notifies the tester by email if applicable.		
Preconditions:	1. PRE-1: The defect report exists and its status is Assigned. 2. PRE-2: The developer is authenticated and is the developer assigned to the defect.		
Postconditions:	1. POST-1: The defect status is updated to Fixed or Cannot reproduce. 2. POST-2: The status change is recorded with date/time (audit/history). 3. POST-3: The system sends an email notification to the tester to inform the status change if applicable.		
Basic Flow:	1. Developer views an Assigned defect report detail. 2. Developer investigates and reproduces the defect. 3. Developer implements a correction. 4. Developer updates the defect status to Fixed. 5. System records the status change and timestamp and displays the updated status. 6. System sends a notification email to the tester if an email address is provided.		
Alternative Flows:	3a. Cannot reproduce the defect: 1. Developer updates the defect status to Cannot Reproduce. 2. System records the status change and timestamp and displays the updated status. 3. System sends a notification email to the tester if an email address is provided.		

	4. Use case ends.
--	-------------------

UC ID and Name:	BT-UC6 Project Owner Retest Defect and Finalize Outcome (Resolved or Reopened)		
Primary Actor:	Project Owner	Supporting Actors:	User Authorization System, Email Service
Trigger:	A defect report is in Fixed status and is ready for retesting/verification		
Description:	The Product Owner retests a defect that a developer marked as Fixed. If the fix is verified successfully, the Product Owner updates the status to Resolved. If retesting fails, the Product Owner updates the status to Reopened and the defect returns to the backlog for further work. BetaTrax records the status change and notifies the tester by email if applicable.		
Preconditions:	- PRE-1: The defect report exists and its status is Fixed. - PRE-2: The Product Owner is authenticated and is the Product Owner of the product.		
Postconditions:	- POST-1: The defect status is updated to Resolved or Reopened. - POST-2: The status change is recorded with date/time (audit/history). - POST-3: The system sends an email notification to the tester to inform the status change if applicable.		
Basic Flow:	- Product Owner opens a Fixed defect report. - Product Owner retests the fix based on the defect's reproduction steps (and any updated notes). - Product Owner updates the defect status to Resolved if retest is successful. - System records the status change and timestamp and displays the updated status. - System sends a notification email to the tester if an email address is provided.		
Alternative Flows:	3a. Retest fails (defect not corrected): - Product Owner updates the defect status to Reopened. - System records the status change and timestamp and displays the updated status. - System sends a notification email to the tester if applicable. - Use case ends (defect is now available for rework via backlog workflow).		

UC ID and Name:	BT-UC7 Add Comment to Defect Report		
Primary Actor:	Product Owner & Developer	Supporting Actors:	User Authorization System
Trigger:	The Developer or Product Owner needs to add detail or record a decision regarding the defect.		
Description:	The Developer or Product Owner attaches a text comment to the defect report at any point during its lifecycle to maintain a history of decisions.		
Preconditions:	- PRE-1: The defect report exists in the system. - PRE-2: The user is authenticated as the Product Owner or Developer.		
Postconditions:	- POST-1: The comment is saved and attached to the report. - POST-2: The comment text, current date, and author identity are permanently recorded.		
Basic Flow:	- The Developer or Product Owner enters comment text on the defect report. - The system attaches the comment to the report.		

	3. The system saves the comment with the text, the date it was attached, and identifies the author.
Alternative Flows:	1a. Empty Comment Text 1. The Developer or Product Owner submits the comment, but the comment text is empty or missing. 2. The system signals an error indicating that comment text is required. 3. The user either provides the text and resubmits, or cancels the operation. 1b. The user decides to cancel the comment 1. The user cancels the operation on the system. 2. The system discards any entered text and the use case ends